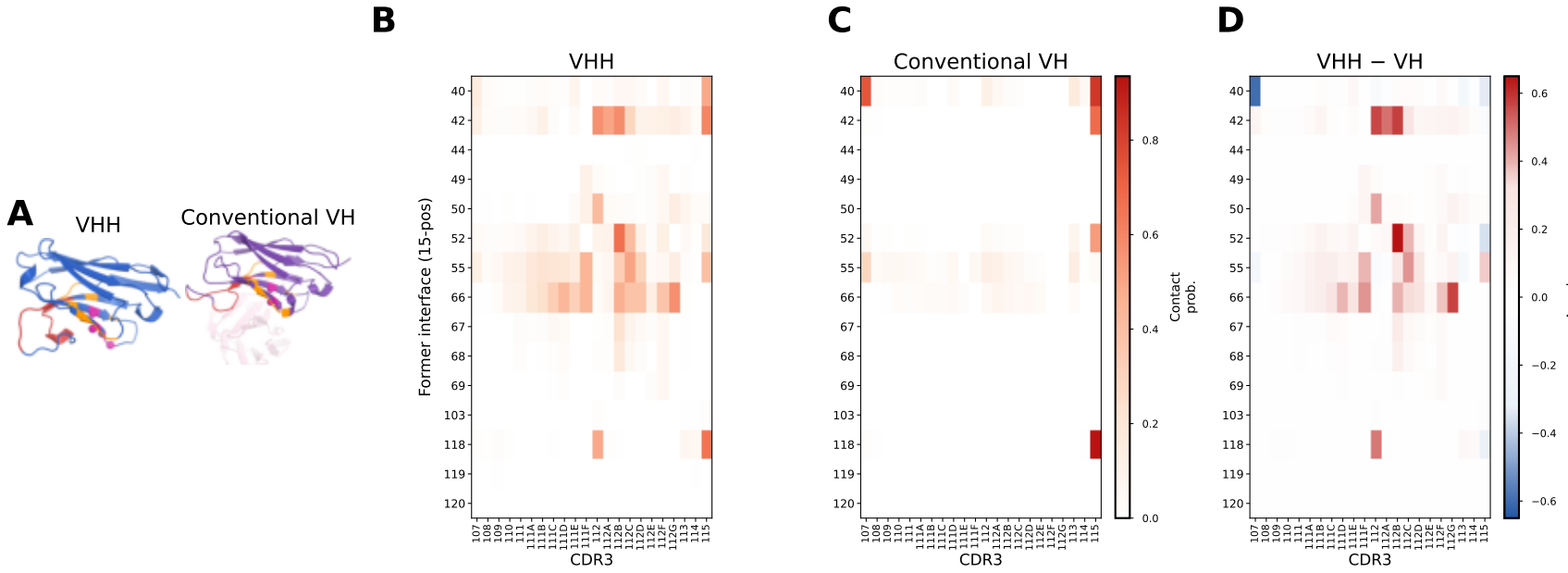


E

Group	Frac. 15-pos contacted	N positions contacted	Min CDR3-IF dist. (Å)	Total contact burden	FR2 contact burden	Non-FR2 burden	Frac. contacts in FR2	FR2_CDR3_Dis (Å)
VHH	0.341 ****	5.12 ****	3.46 ****	9.90 ****	6.87 ****	3.03 ****	0.719 ****	7.95 ****
Conventional VH	0.267 ****	4.01 ****	2.97 ****	5.44 ****	4.26 ****	1.18 ****	0.773 ****	8.66 ****

FIGURE 4A

CDR3-former-interface geometry. (A) Representative VHH (blue; PDB pdb_00008qf4-E) and conventional VH (purple; PDB pdb_00009f1i-H-L) domains; CDR3=red, 15-position former-interface target=orange, canonical tetrad (IMGT 42/49/50/52)=magenta spheres. (B,C) CDR3-to-former-interface minimum-distance contact probability (≤4.5Å) for VHH (B) and conventional VH (C), identical color scale. (D) VHH−VH differential contact probability. (E) Cohort-level geometry summary (mean; VHH N=533, VH N=1428-1429); significance = BH-FDR q (ns/*/**/***/**** = q≥0.05/<0.05/<0.01/<0.001/<0.0001).